

# Ch 2 – Quadratic Functions

## Finding vertex/stationary point:

Given the curve  $f(x) = ax^2 + bx + c$ , the stationary point/vertex can be found by either completing the square or by differentiation

Completing the square:

$$\begin{aligned} & ax^2 + bx + c \\ &= a \left[ x^2 + \frac{b}{a}x \right] + c \\ &= a \left[ \left( x + \frac{b}{2a} \right)^2 - \left( \frac{b}{2a} \right)^2 \right] + c \\ &= a \left[ \left( x + \frac{b}{2a} \right)^2 - \frac{b^2}{4a^2} \right] + c \\ &= a \left( x + \frac{b}{2a} \right)^2 - \frac{b^2}{4a} + c \end{aligned}$$

If  $a > 0$ , the curve has a minimum

If  $a < 0$ , the curve has a maximum

$$\text{Coordinates} = \left( \frac{-b}{2a}, c - \frac{b^2}{4a} \right)$$

Differentiation:

$$y = ax^2 + bx + c$$

$$\frac{dy}{dx} = 2ax + b$$

$$2ax + b = 0$$

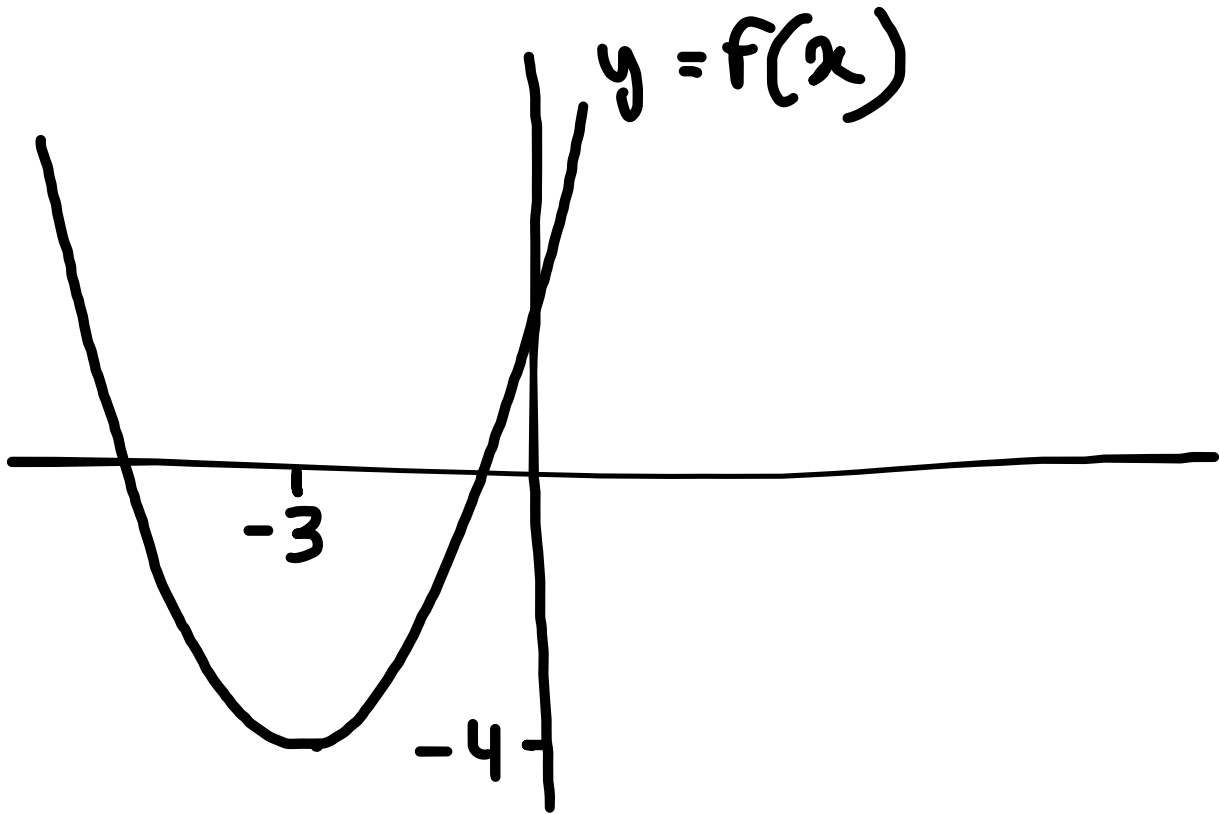
$$x = \frac{-b}{2a}$$

$$y = a \left( \frac{-b}{2a} \right)^2 + b \left( \frac{-b}{2a} \right) + c$$

## Sketching quadratic functions:

Eg:  $f(x) = x^2 + 6x + 5$

Vertex =  $(-3, -4)$



## Determining range:

Eg:  $f(x) = x^2 + 6x + 5$ , for  $x \geq -3$

$$f \geq -4$$

## Number of roots/Number of intersections with a line:

For  $f(x) = ax^2 + bx + c$

Discriminant  $= b^2 - 4ac$

If  $b^2 - 4ac > 0$ , the function has two real roots/intersects the line twice

If  $b^2 - 4ac = 0$ , the function has two equal roots/intersects the line once (line is a tangent)

If  $b^2 - 4ac < 0$ , the function has no real roots/does not intersect the line

## Solving quadratic equations:

Factorisation:

Eg:  $x^2 + 5x + 6 = 0$

$$6 = 2 \times 3$$

$$5 = 2 + 3$$

$$(x + 2)(x + 3) = 0$$

$$x = -3, x = -2$$

Quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Eg:  $x^2 + 5x + 6 = 0$

$$x = \frac{-5 \pm \sqrt{25 - 24}}{2} = \frac{-5 \pm 1}{2}$$

$$x = -3, x = -2$$

Completing the square:

Eg:  $x^2 + 5x + 6 = 0$

$$\left(x + \frac{5}{2}\right)^2 - \left(\frac{5}{2}\right)^2 + 6 = 0$$

$$\left(x + \frac{5}{2}\right)^2 = \frac{1}{4}$$

$$x = -3, x = -2$$

## Solving quadratic inequalities:

$$x^2 + 11x + 10 > 0$$

$$(x + 10)(x + 1) > 0$$

$$x < -10, x > -1$$



$$x^2 + 6x + 8 < 0$$

$$(x + 4)(x + 2) < 0$$

$$-4 < x < -2$$

